A rank-one lattice with unbounded stable commutator length

Counterexample to problem 00000001158

lizaixi01

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Scope and result. The printed conjecture asserts that a lattice has bounded stable commutator length if and only if its rank is one, with a knot-complement exception. Use the usual ambient real-rank convention for lattices in semisimple real Lie groups. Put

$$\overline{\Gamma(2)} = \Gamma(2)/\{\pm I\} < \mathrm{PSL}_2(\mathbb{R}), \qquad \Gamma(2) = \ker\big(\mathrm{SL}_2(\mathbb{Z}) \longrightarrow \mathrm{SL}_2(\mathbb{F}_2)\big).$$

This standard nonuniform lattice has ambient real rank one and is a free group $F = \langle a, b\rangle$ of rank two. For $c = [a, b]$, the supplied Lean proof establishes, for every integer $N \geq 0$,

$$\frac{N}{12} \;\leq\; \mathrm{scl}_F(c^N) \;\leq\; N. \tag{1}$$

Thus the actual finite values on the commutator subgroup are unbounded. The example is not the group of a usual single knot in S^3: its abelianization has a noncyclic quotient $\mathbb{Z}^2$, whereas such knot groups have cyclic abelianization. The conjecture prints no cocompactness restriction. No extended-infinite value outside $[F, F]$ is used.

Published geometric identification. The exact geometric and topological identifications in this paragraph are used from published standard models; they are *not re-proved in Lean.* Morris, Example 1.3.7, proves that $\mathrm{SL}_2(\mathbb{Z})$ has finite Haar covolume in $\mathrm{SL}_2(\mathbb{R})$ by integrating over its modular fundamental domain. Corollary 4.1.14(2) preserves the lattice property under finite index. The kernel defining $\Gamma(2)$ has finite index, as Conrad explains before Theorem 3.1. Quotienting both groups by the finite central subgroup $\{\pm I\}$ preserves discreteness and the finite invariant quotient measure, since the corresponding lattice quotient spaces agree. Morris, Example 8.1.7(1) and Remark 8.1.9(2), give real rank $\mathrm{rank}_{\mathbb{R}}\,\overline{\mathrm{SL}_2(\mathbb{R})} = 1$ and invariance under compact central quotients. Conrad, Remark 3.10 (pp.10–11), identifies $\overline{\Gamma(2)}$ with a free group of rank two. Importantly, $\Gamma(2)$ itself contains $-I$ and is not that free group. Burde–Zieschang, Chapter 8A, p.103, gives the cyclic abelianization of classical knot groups. The Lean theorem transports the algebraic result through every group isomorphism; no geometric axiom or assigned lattice or rank label is introduced.

Genuine minimum and genuine stable limit. For an arbitrary group G and a finite list $L = ((u_i, v_i))$, define its actual product $P(L) = \prod_i [u_i, v_i]$. The proof shows that all such products form precisely Mathlib's native subgroup $[G, G]$; inverse products are represented by reversing the list and swapping the pairs. For every $x \in [G, G]$ the minimum

$$\mathrm{cl}_G(x) = \min\{|L| : P(L) = x\}$$

therefore exists. It is implemented by `Nat.find` only after this existence proof, with an actual minimizing witness and the inequality against *every* representing list. Concatenation proves $\mathrm{cl}_G(xy) \leq \mathrm{cl}_G(x) + \mathrm{cl}_G(y)$. The sequence $u_n = \mathrm{cl}_G(x^n)$ is subadditive, and $u_n/n \geq 0$. Mathlib's native Fekete theorem then proves the genuine limit

$$\mathrm{scl}_G(x) = \lim_{n\to\infty} \frac{\mathrm{cl}_G(x^n)}{n} = \inf_{n\geq 1} \frac{\mathrm{cl}_G(x^n)}{n}, \qquad 0 \leq \mathrm{scl}_G(x) \leq \mathrm{cl}_G(x).$$

The nonempty, bounded-below sequence supplies the infimum and its limit; neither a desired norm nor a fallback infimum value is assigned.

Actual Brooks count and its defect. For the native free group on two generators, let $w(g)$ be its actual reduced word. Write $A = a^{-1}$ and $B = b^{-1}$. The integer

$$H(g) = \#\{ab \text{ occurrences in } w(g)\} - \#\{BA \text{ occurrences in } w(g)\}$$

is constructed from adjacent-letter indicators. The proof identifies the recursive sum with actual `List.zip`, `tail` and `countP` occurrence counts. It proves $H(g^{-1}) = -H(g)$. For any two reduced words the proof

constructs the full cancellation decomposition $w(g) = UC$, $w(h) = C^{-1}V$, with UV reduced. Each of the three remaining word junctions changes the signed count by at most one. Hence

$$|H(gh) - H(g) - H(h)| \le 3, \qquad |H([g,h])| \le 9.$$

Induction on every commutator-product list gives $|H(P(L))| \le 12|L|$. These are proved bounds for the concrete count, consistent with the standard counting construction in Calegari, Definition 2.25 and Lemma 2.29; no quasimorphism is assumed.

Unbounded finite stable lengths. The native word of c is $abAB$. It is cyclically reduced, and the proof identifies the exact reduced word of every c^n with the concatenation of n copies of this word. Its actual occurrence count is $H(c^n) = n$. Consequently, for every positive n and every $N \ge 0$,

$$Nn = H((c^N)^n) \le 12\,\mathrm{cl}_F((c^N)^n), \qquad \frac{N}{12} \le \frac{\mathrm{cl}_F((c^N)^n)}{n}.$$

Passing to the genuine stable limit proves the lower bound in (1). A list of N copies of the pair (a,b) represents c^N, proving the upper bound. For every real C, choose a natural $N > 12C$; then $C < \mathrm{scl}_F(c^N) \le N$. Lean proves this quantified witness statement and the failure of boundedness of the full range. It also proves that genuine commutator lengths and stable limits are preserved by every group isomorphism, by transporting all product lists in both directions and using uniqueness of limits.

The knot exception is excluded. The genuine exponent-sum homomorphism $F \to (\mathbb{Z}^2, +)$ sends a to $(1,0)$ and b to $(0,1)$. It is onto: $a^m b^n$ maps to (m,n) for all integers m, n. Mathlib's universal map through the native `Abelianization` therefore gives a surjection $F^{\mathrm{ab}} \twoheadrightarrow \mathbb{Z}^2$. The proof rules out a cyclic generator (r,s) of $\mathbb{Z}^2$: representing $(1,0)$ forces $s = 0$, making $(0,1)$ impossible. Thus F^{ab} is not cyclic. A final theorem excludes every group isomorphism from F to any group with cyclic native abelianization, which applies to the published knot-group theorem above. This argument does not address arbitrary link complements or knots in other ambient manifolds.

Formalization and reproducibility. The complete algebraic proof is in `lean/Main.lean`, namespace `TLMC1158`, culminating in `complete_counterexample`. The geometric model identification, Haar covolume and Lie rank, and the general knot-group theorem are the explicitly disclosed external boundaries. The proof neither formalizes these identifications by an extra axiom nor claims to prove a general classification of lattices or knot groups. The target is only the printed rank-one boundedness implication under the stated conventions.

The project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. Run `lean --version`, then `lean/verify.sh` for the build, ordinary exhaustive namespace axiom audit and official fresh-kernel replay. Command logs, original job and resource observations, exact source bindings and the file manifest accompany a verified submission. Production timings do not establish a controlled efficiency gain.

References

[1] D. Calegari, *scl*, MSJ Memoirs 20, Mathematical Society of Japan, 2009, pp.13,18,21–22. https://www.math.uchicago.edu/~dannyc/books/scl/scl.pdf.

[2] K. Conrad, $SL_2(\mathbb{Z})$, Theorem 3.1 and Remark 3.10, pp.7,10–11. https://kconrad.math.uconn.edu/blurbs/grouptheory/SL(2,Z).pdf.

[3] D. W. Morris, *Introduction to Arithmetic Groups*, arXiv:math/0106063v6, 2015, pp.15–16,46,180–181. https://arxiv.org/abs/math/0106063v6.

[4] G. Burde and H. Zieschang, *Knots*, second revised and extended edition, de Gruyter, 2003, Chapter 8A, p.103. https://www.maths.ed.ac.uk/~v1ranick/papers/burdzies.pdf.